R84/D_attention_focal | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 30 | update 43200 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	70.9543	66.2037	11.417064	11.427978	11.371692	-0.1898	+0.3979	0.4081
TEST ALL	71.2173	66.1802	11.373922	11.385299	11.329932	-0.1969	+0.3862	0.3952

Dataset	n	NSS Top1 (%)	R84 Top1 (%)	NSS cost	R84 cost
TSP TEST	4500	45.888889	47.000000	7.970400719	7.971416858
CVRP TEST	4500	45.977778	47.866667	18.004802368	18.013367774
TSPLIB	49	20.408163	28.571429	8.217907074	8.199911482
CVRPLIB	100	70.000000	58.000000	67.101111717	67.299114523

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: direct_attention | focal extra: True

R84/D_attention_focal | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	70.9543	66.2037	11.417064	11.427978	11.371692	-0.1898	+0.3979	0.4081	per-problem TRAIN SBS
TSP	46.8667	33.1111	8.025464	8.045354	7.990533	-0.2472	+0.4372	0.4272	T2T
CVRP	46.7778	18.7111	18.162843	18.269377	18.064087	-0.5831	+0.5467	0.7302	RELD_CVRP
ATSP	77.2889	47.4889	1.638592	1.676905	1.632471	-2.2848	+0.3750	0.3721	UNICO_MatPOENet
OVRP	40.2444	35.7556	7.463110	7.469579	7.415777	-0.0866	+0.6383	0.6488	MoSES_RF
VRPB	81.1111	80.8444	9.357749	9.354876	9.318750	+0.0307	+0.4185	0.4197	reld_mt
VRPL	36.6222	33.0889	12.432514	12.437959	12.383444	-0.0438	+0.3963	0.4023	MoSES_RF
VRPTW	80.0444	79.9556	18.818829	18.819431	18.766139	-0.0032	+0.2808	0.2966	reld_mt
OVRPTW	83.8444	83.7778	10.143009	10.143730	10.116859	-0.0071	+0.2585	0.2558	reld_mt
OVRPB	88.5111	88.5111	6.357277	6.357254	6.341078	+0.0004	+0.2555	0.2668	reld_mt
OVRPL	40.2000	35.9111	7.438174	7.445709	7.390416	-0.1012	+0.6462	0.6607	MoSES_RF
VRPBL	78.8222	78.5111	9.368773	9.371523	9.328366	-0.0293	+0.4332	0.4372	reld_mt
VRPBTW	78.6222	78.4222	19.946456	19.950430	19.851084	-0.0199	+0.4804	0.4595	reld_mt
VRPLTW	78.8444	78.7556	18.764450	18.765871	18.707843	-0.0076	+0.3026	0.3163	reld_mt
OVRPBL	89.2667	89.2000	6.370021	6.370093	6.352824	-0.0011	+0.2707	0.2860	reld_mt
OVRPBTW	83.6222	83.4444	10.660215	10.661098	10.623245	-0.0083	+0.3480	0.3209	reld_mt
OVRPLTW	83.5778	83.4889	10.157210	10.157423	10.129818	-0.0021	+0.2704	0.2755	reld_mt
VRPBLTW	79.6667	79.4000	19.792239	19.797044	19.706420	-0.0243	+0.4355	0.4242	reld_mt
OVRPBLTW	83.2444	83.2889	10.610225	10.609955	10.571302	+0.0025	+0.3682	0.3460	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R84/D_attention_focal | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2173	66.1802	11.373922	11.385299	11.329932	-0.1969	+0.3862	0.3952	per-problem TRAIN SBS
TSP	47.0000	34.5333	7.971417	7.989621	7.937593	-0.2278	+0.4261	0.4160	T2T
CVRP	47.8667	17.9333	18.013368	18.124983	17.918509	-0.6158	+0.5294	0.6966	RELD_CVRP
ATSP	77.0000	47.2444	1.635222	1.674207	1.629441	-2.3286	+0.3547	0.3544	UNICO_MatPOENet
OVRP	40.1111	33.6444	7.446093	7.455750	7.400487	-0.1295	+0.6162	0.6262	MoSES_RF
VRPB	80.2222	80.4000	9.353003	9.351810	9.315298	+0.0128	+0.4048	0.4132	reld_mt
VRPL	37.1111	32.2667	12.349534	12.356138	12.300958	-0.0534	+0.3949	0.4030	MoSES_RF
VRPTW	81.1556	80.8444	18.688414	18.690545	18.637408	-0.0114	+0.2737	0.2849	reld_mt
OVRPTW	83.9333	83.9333	10.092533	10.093063	10.068135	-0.0052	+0.2423	0.2335	reld_mt
OVRPB	88.0889	88.1333	6.370377	6.370826	6.352225	-0.0070	+0.2858	0.2950	reld_mt
OVRPL	41.0667	34.6000	7.463823	7.472542	7.418033	-0.1167	+0.6173	0.6242	MoSES_RF
VRPBL	79.4667	79.4222	9.394599	9.396379	9.355967	-0.0189	+0.4129	0.4103	reld_mt
VRPBTW	78.8000	78.7556	19.665724	19.665725	19.577666	-0.0000	+0.4498	0.4492	reld_mt
VRPLTW	79.2444	79.2444	18.802106	18.803136	18.745143	-0.0055	+0.3039	0.3226	reld_mt
OVRPBL	89.2444	89.2889	6.375824	6.375741	6.359310	+0.0013	+0.2597	0.2714	reld_mt
OVRPBTW	84.9333	84.8222	10.659407	10.659591	10.624748	-0.0017	+0.3262	0.2985	reld_mt
OVRPLTW	83.0222	82.8667	10.085520	10.086112	10.059001	-0.0059	+0.2636	0.2649	reld_mt
VRPBLTW	79.9556	79.7333	19.702124	19.707068	19.613284	-0.0251	+0.4530	0.4344	reld_mt
OVRPBLTW	83.6889	83.5778	10.661509	10.662149	10.625574	-0.0060	+0.3382	0.3144	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R84/D_attention_focal | LIB and locked provenance

Current R84 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	28.5714	10.2041	8.199911	8.240435	8.142167	-0.4918	+0.7092	0.6669
CVRPLIB	58.0000	45.0000	67.299115	67.245364	66.945093	+0.0799	+0.5288	0.6552

SHA256: addbfaff33b5720dfec357a4b5685e75e161deea62d7bde04b15f48fb39b3efe

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, direct attention logits, stable ascending-global-solver-ID tie break

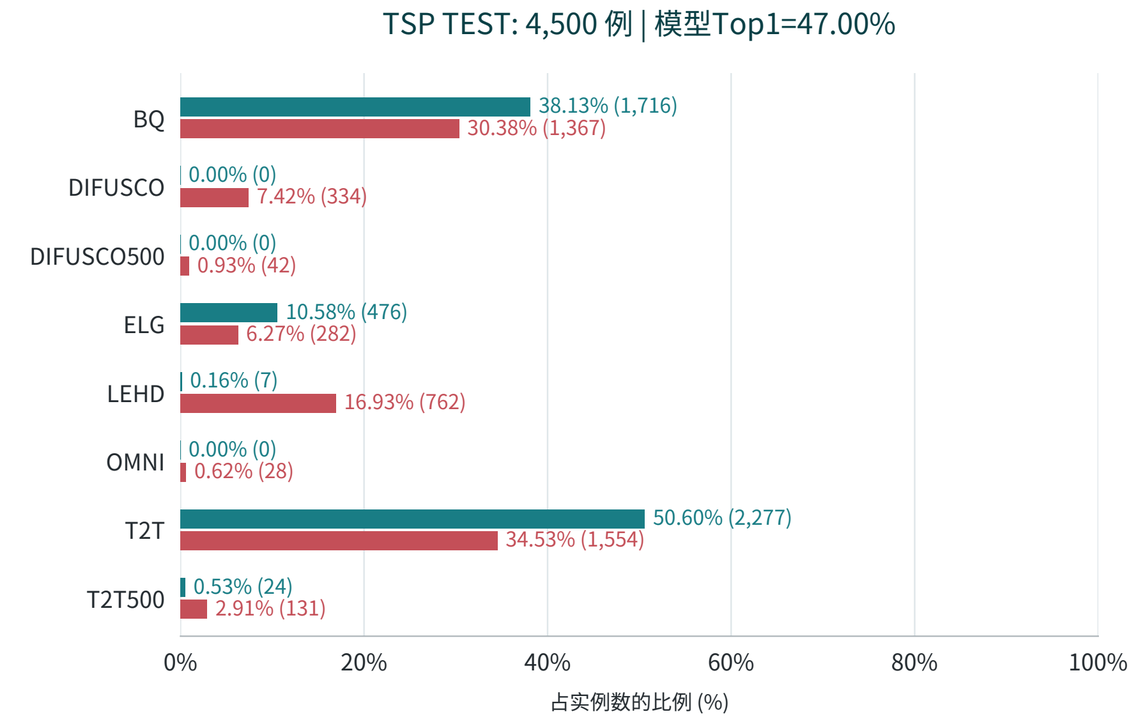
Normalization: new81000 TRAIN only; no TEST/LIB refit.

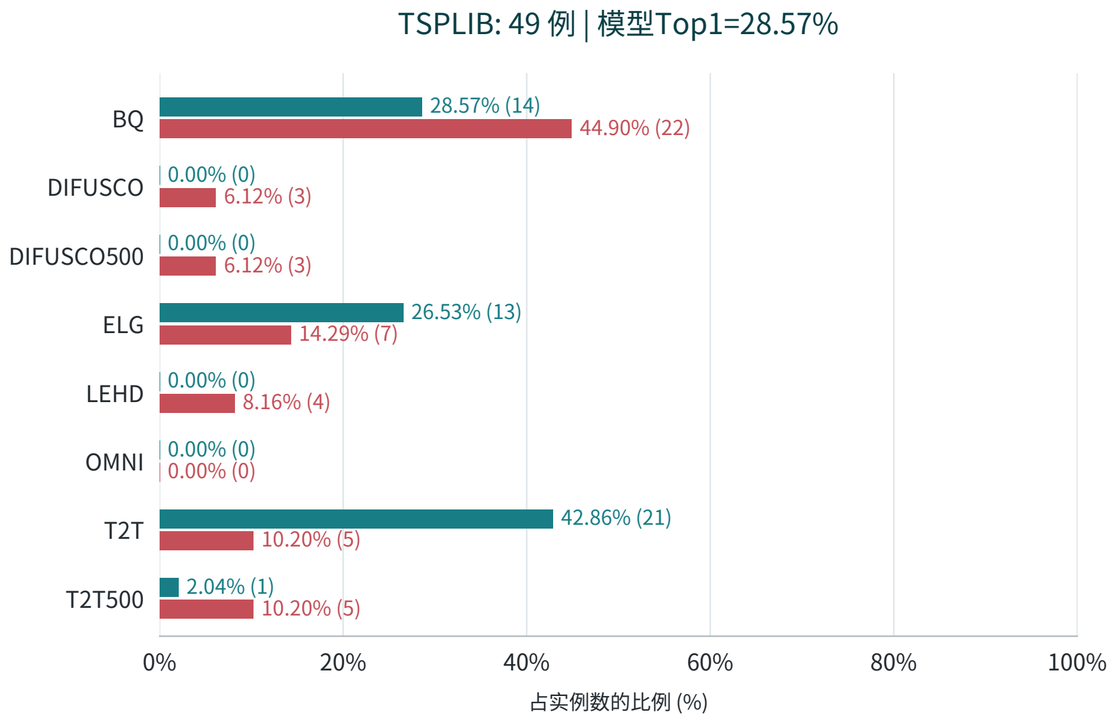
R84/D_attention_focal | TSP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

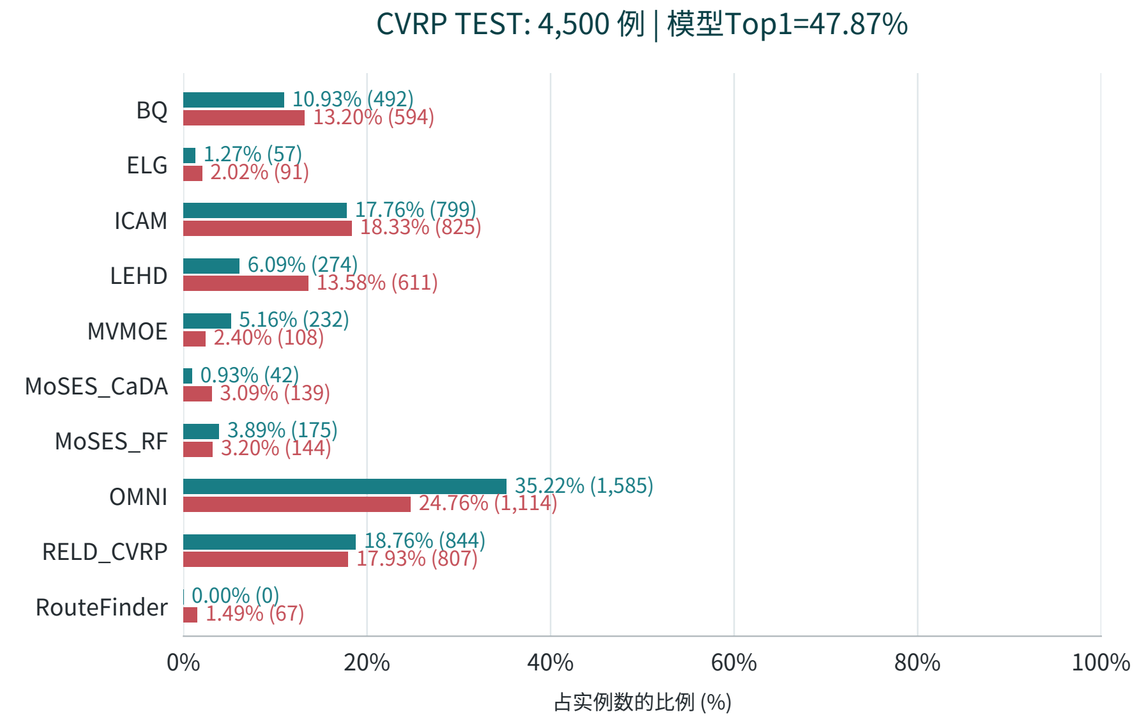
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

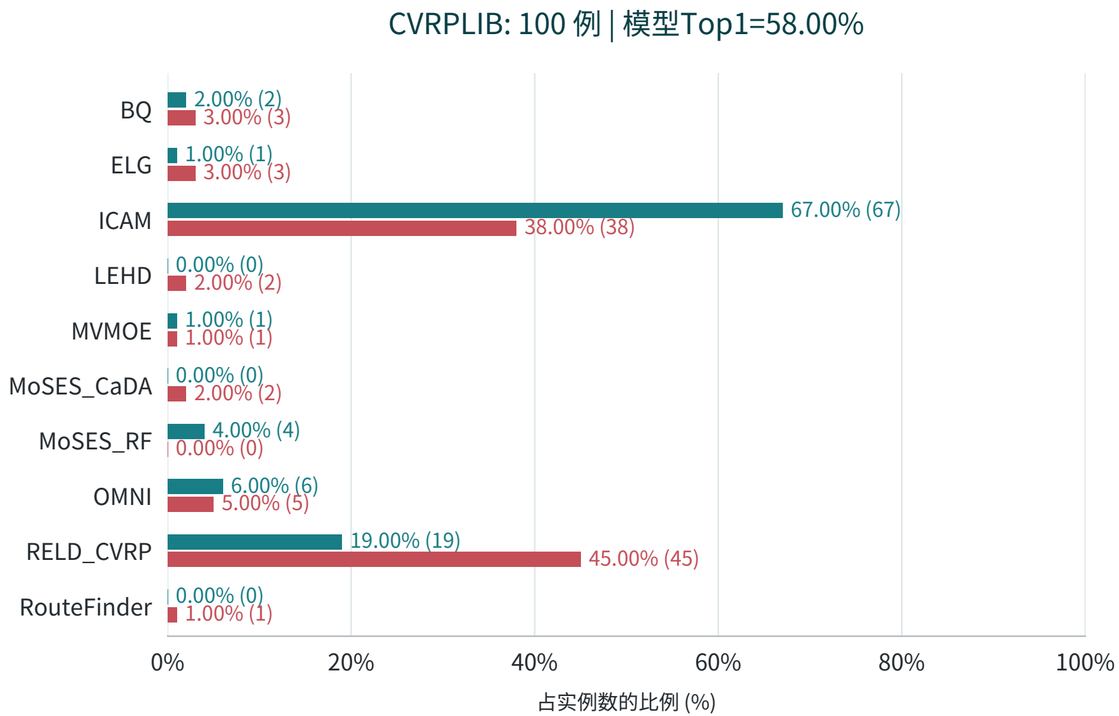
R84/D_attention_focal | CVRP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

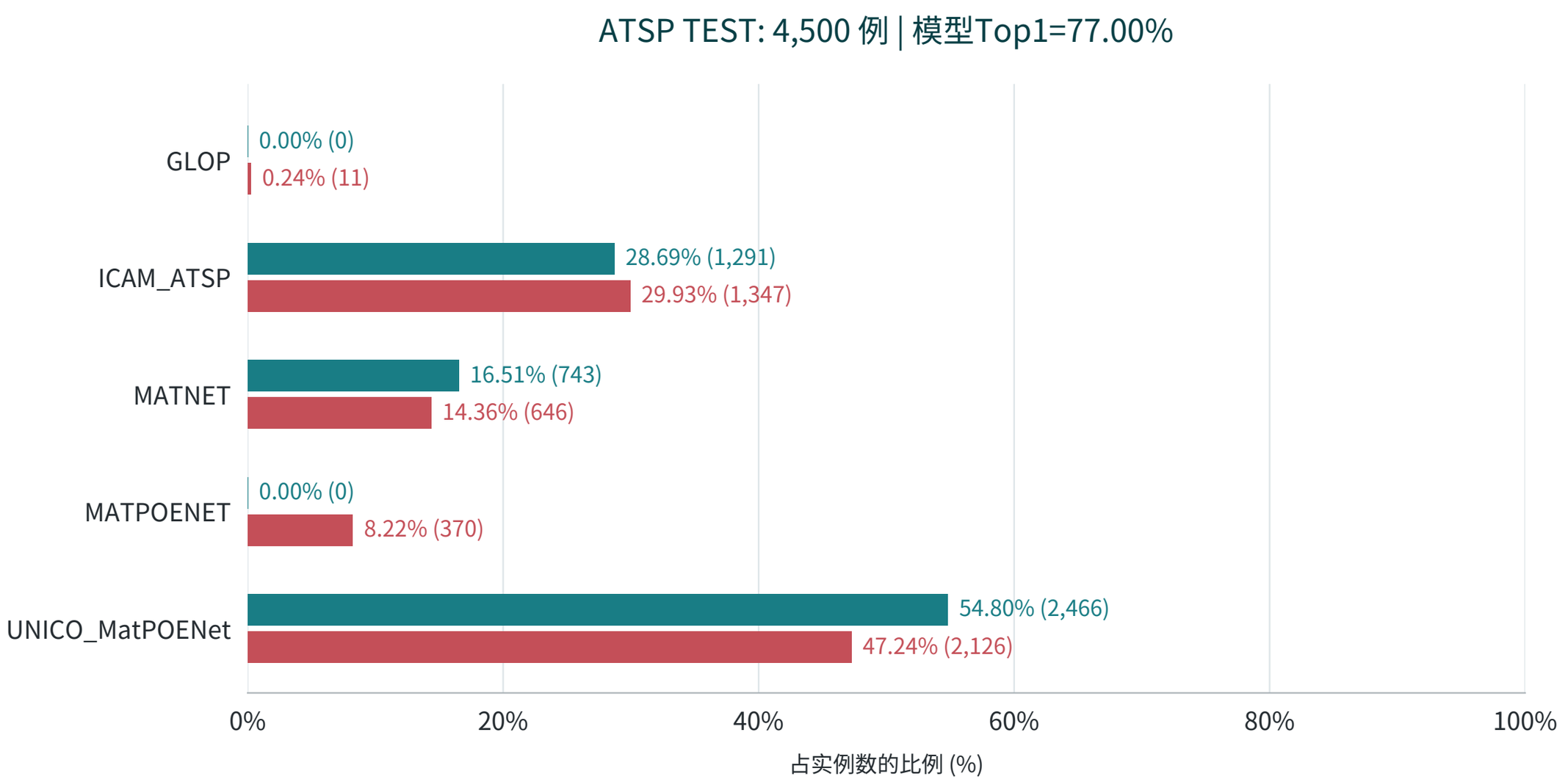
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | ATSP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

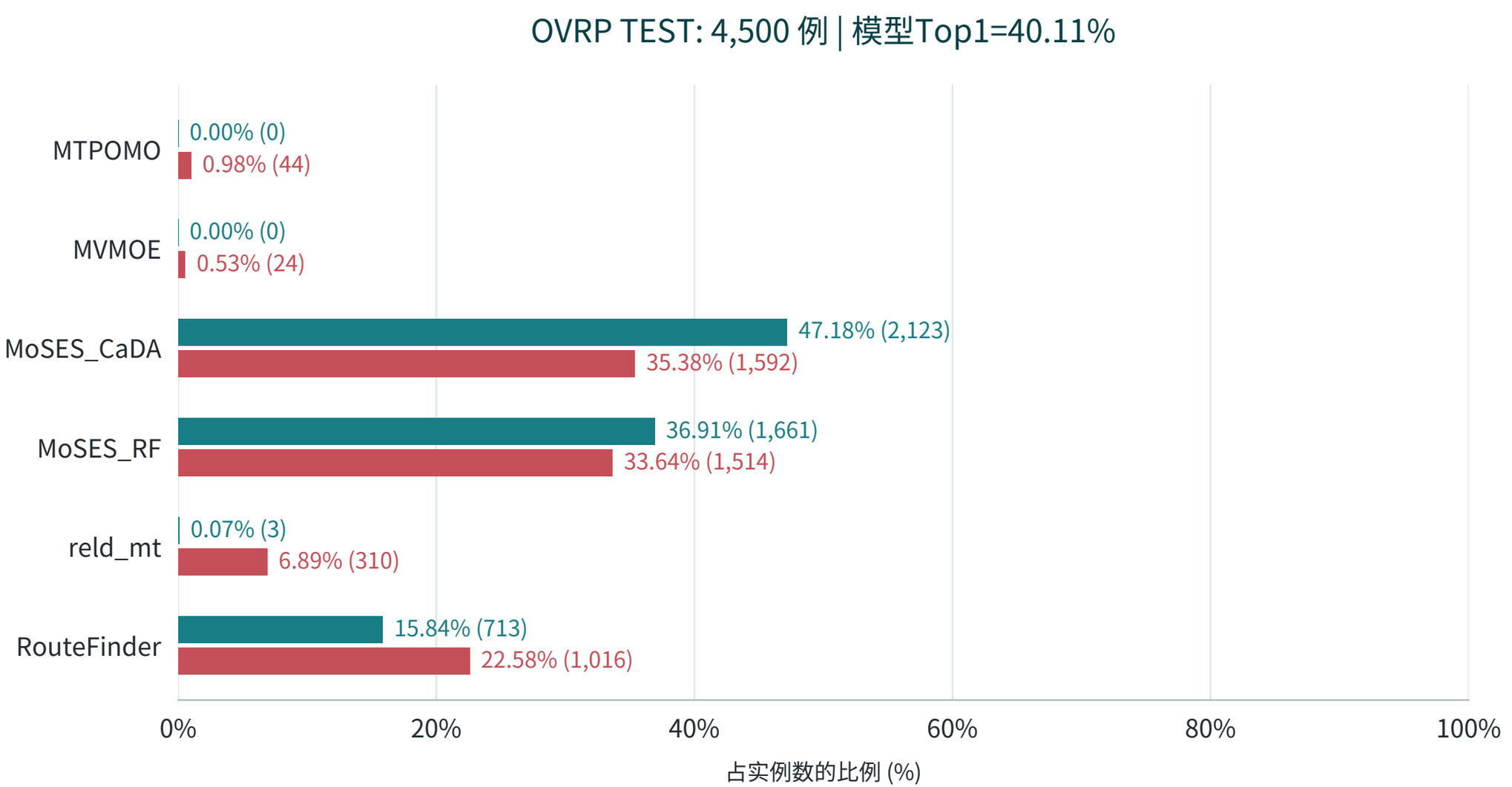
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRP | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

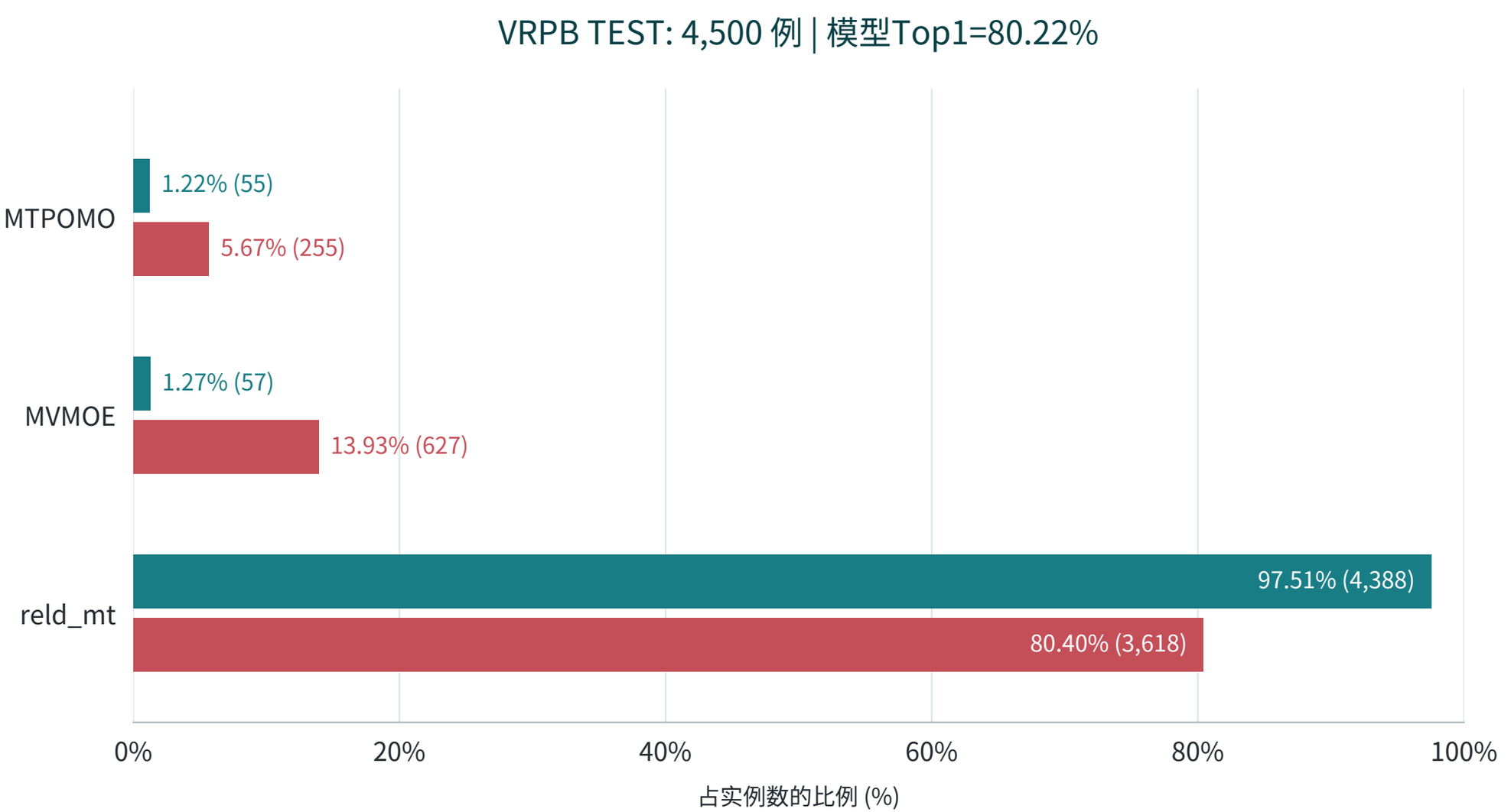
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPB | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

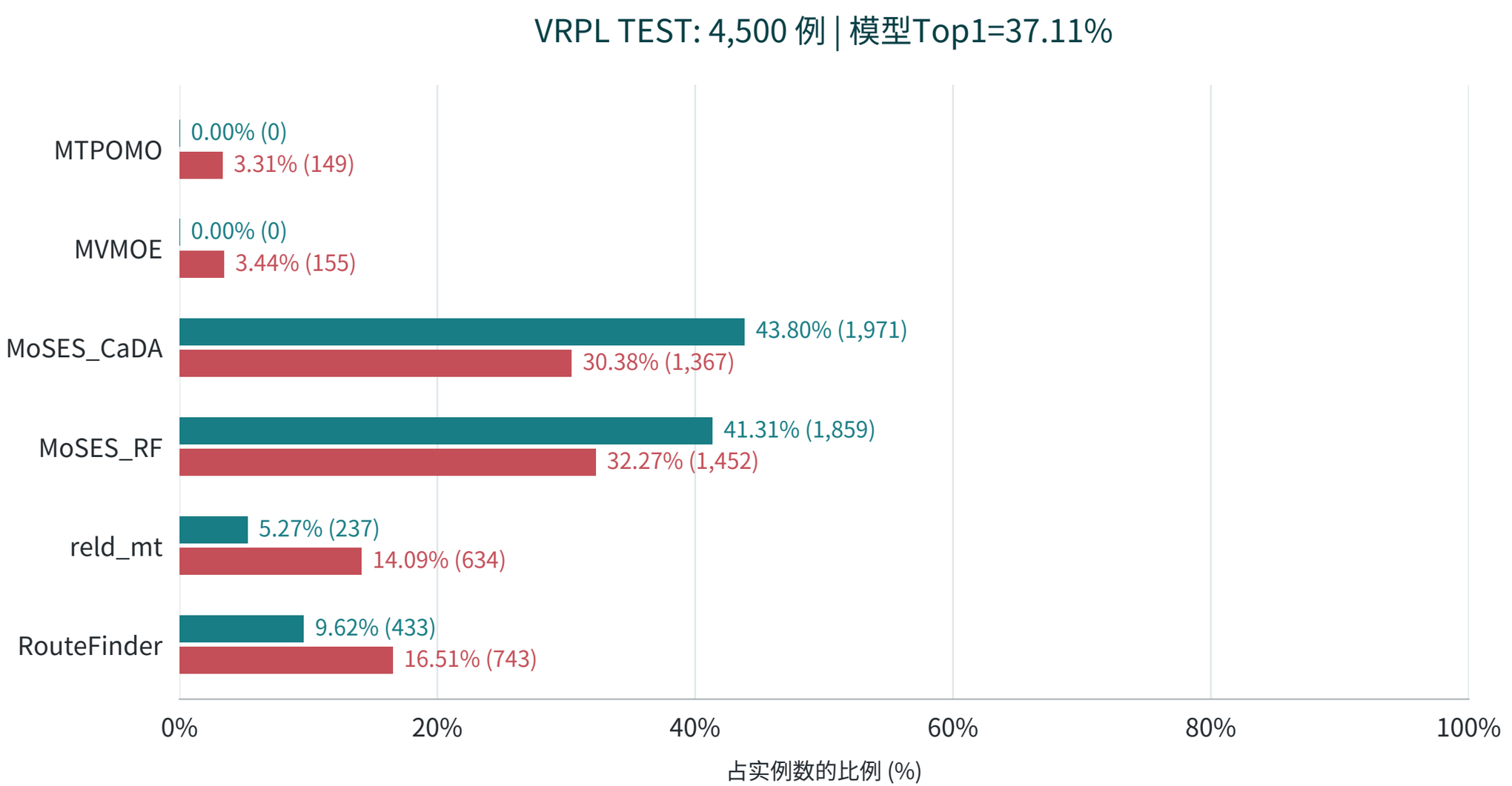
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

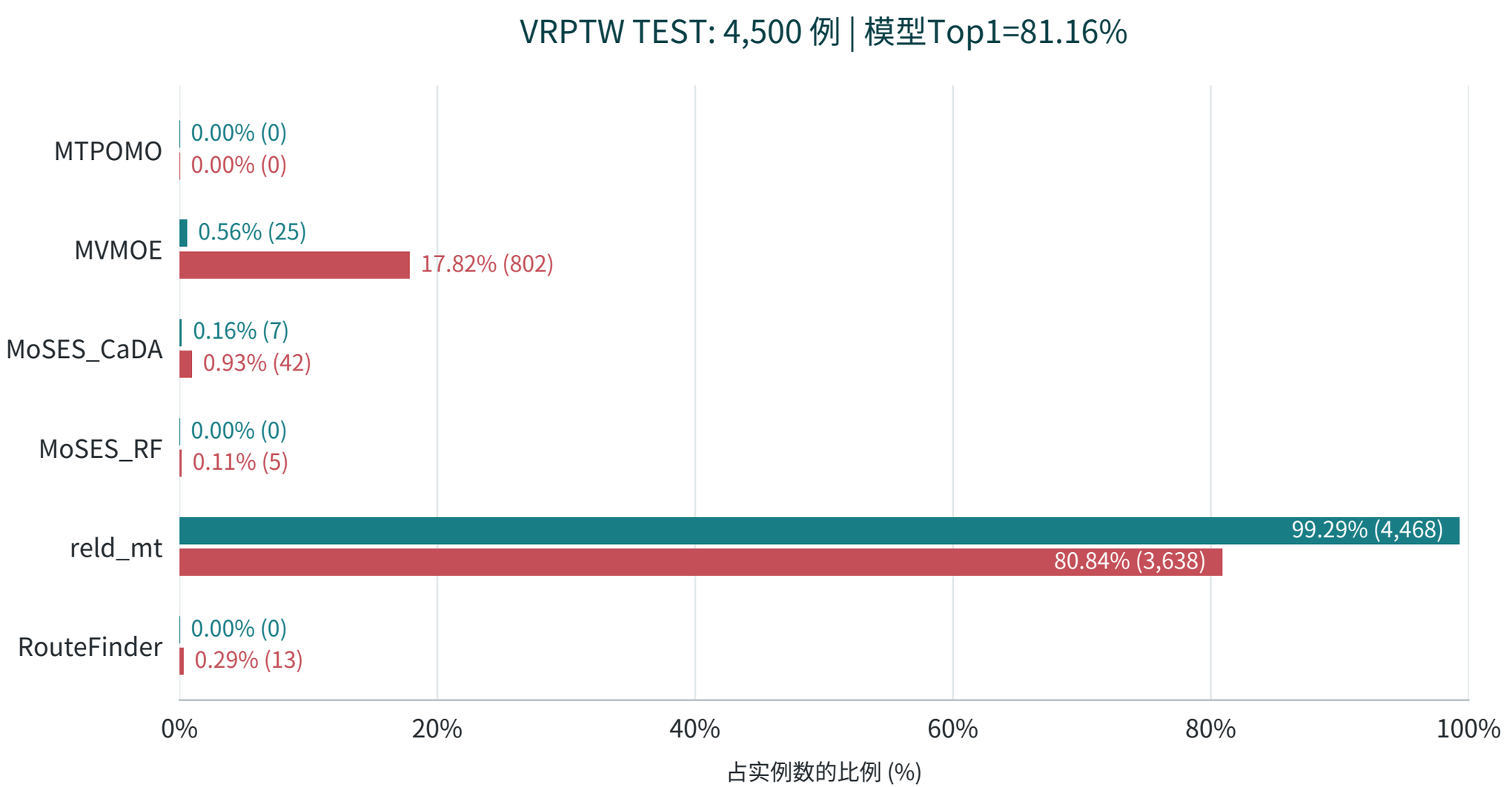
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

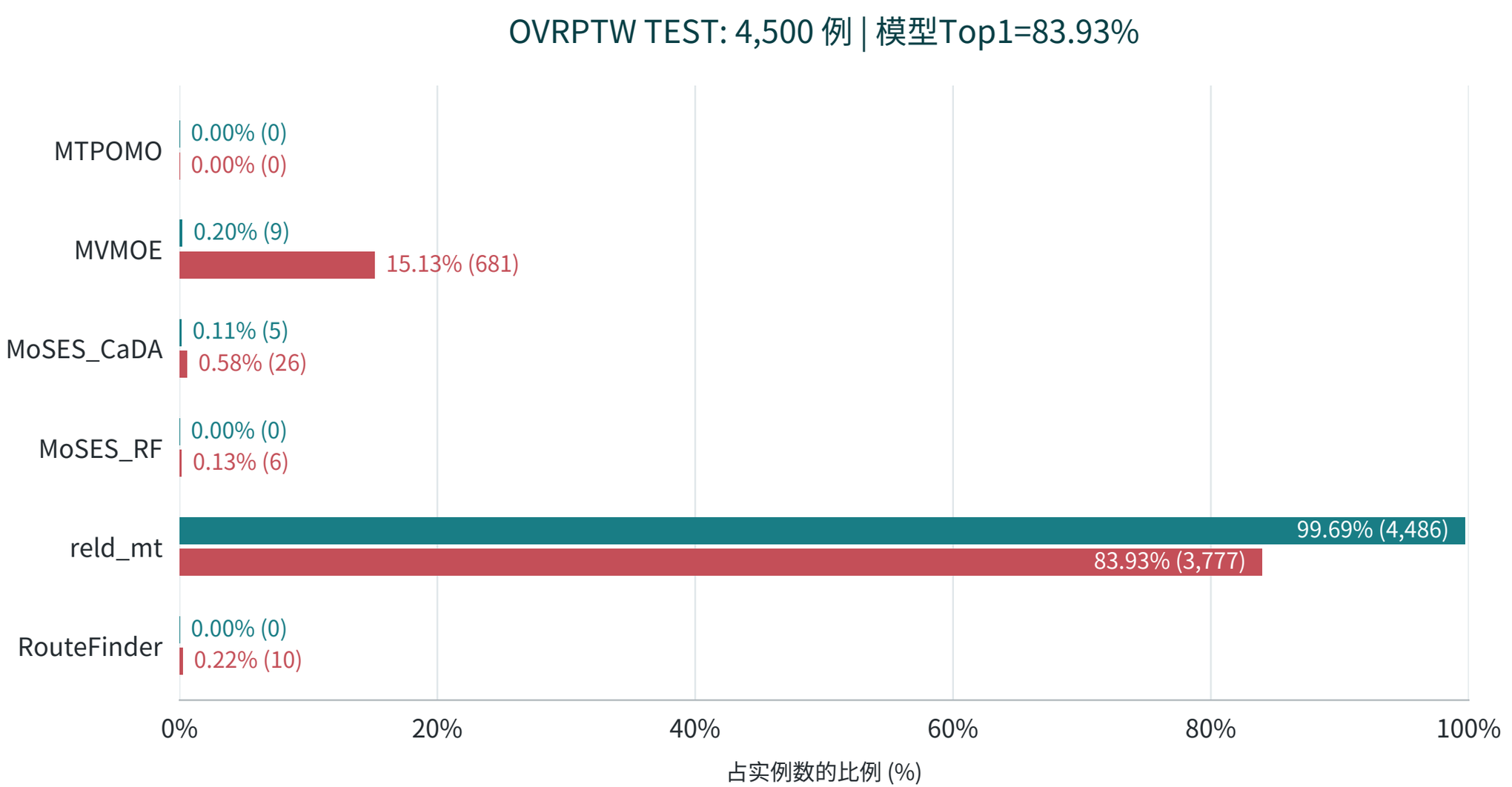
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

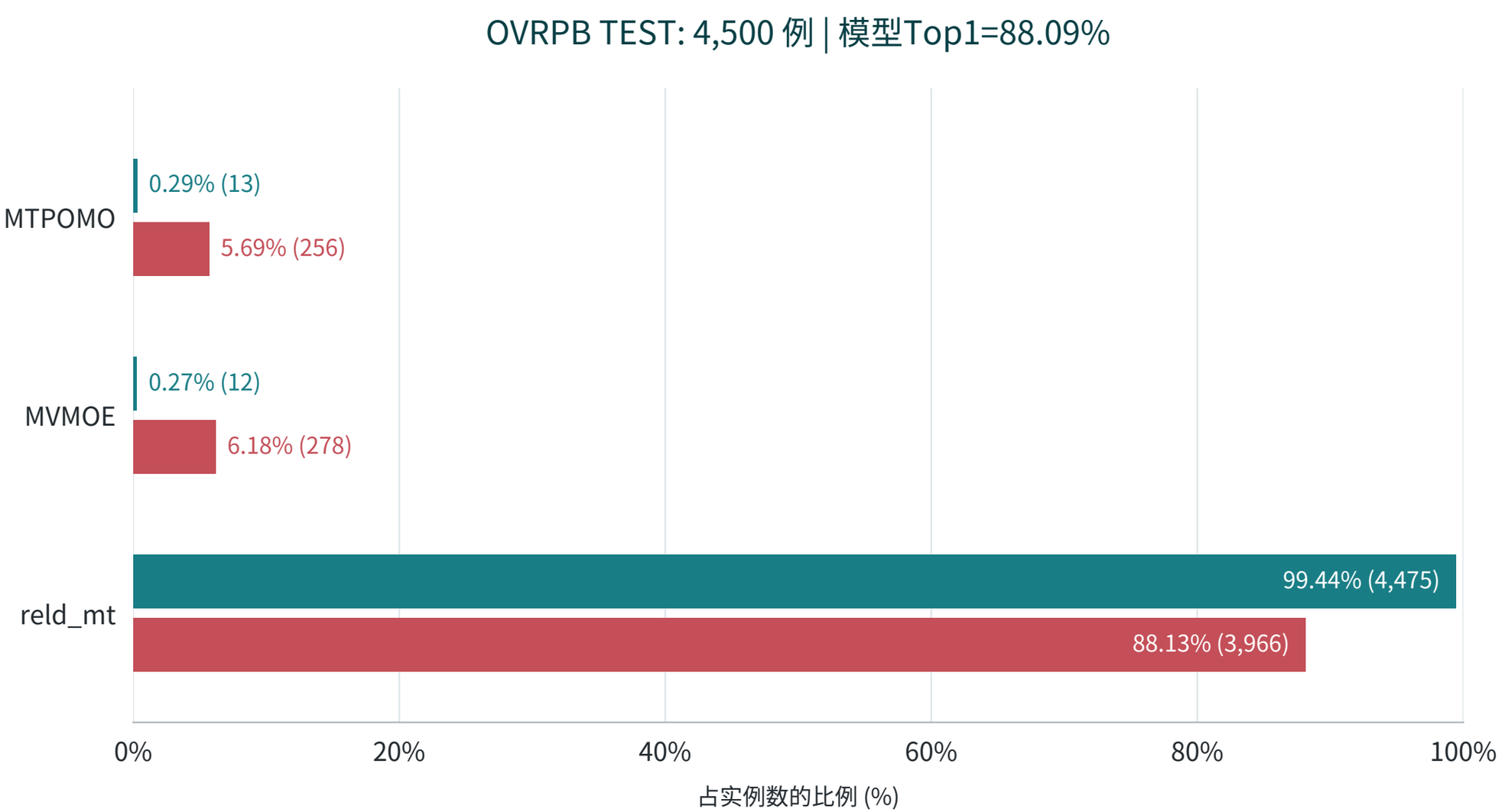
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPB | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

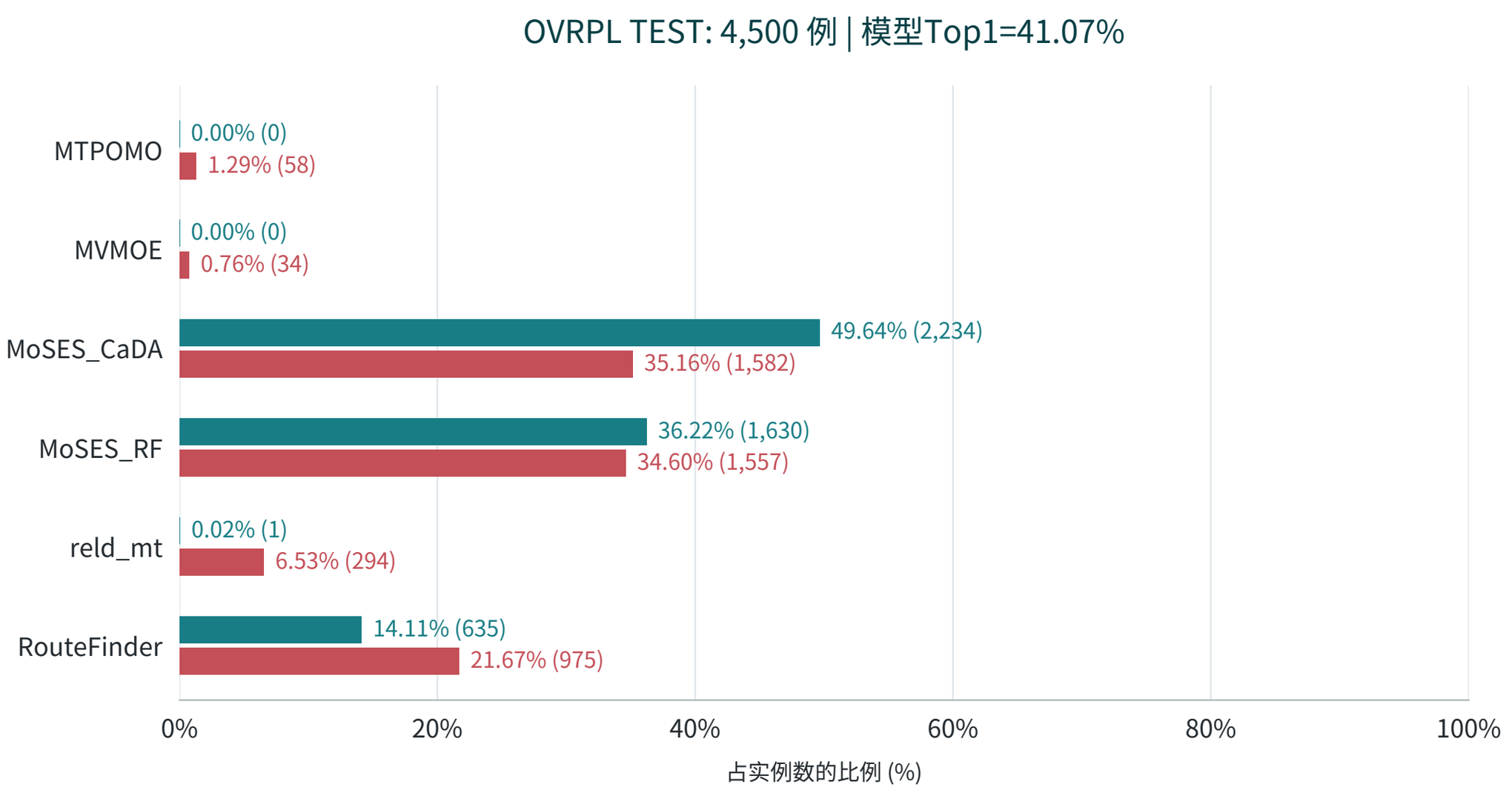
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

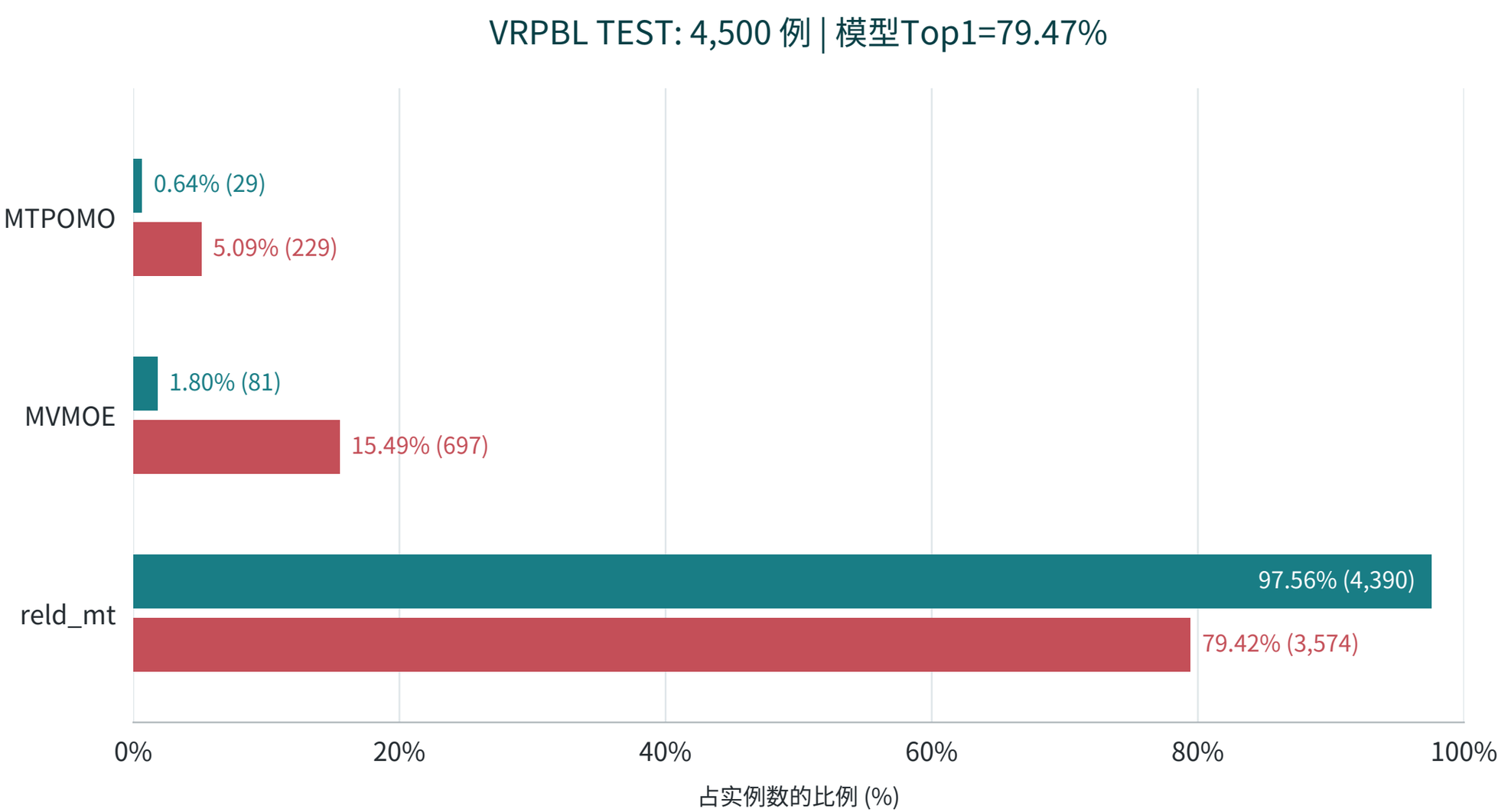
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPBL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

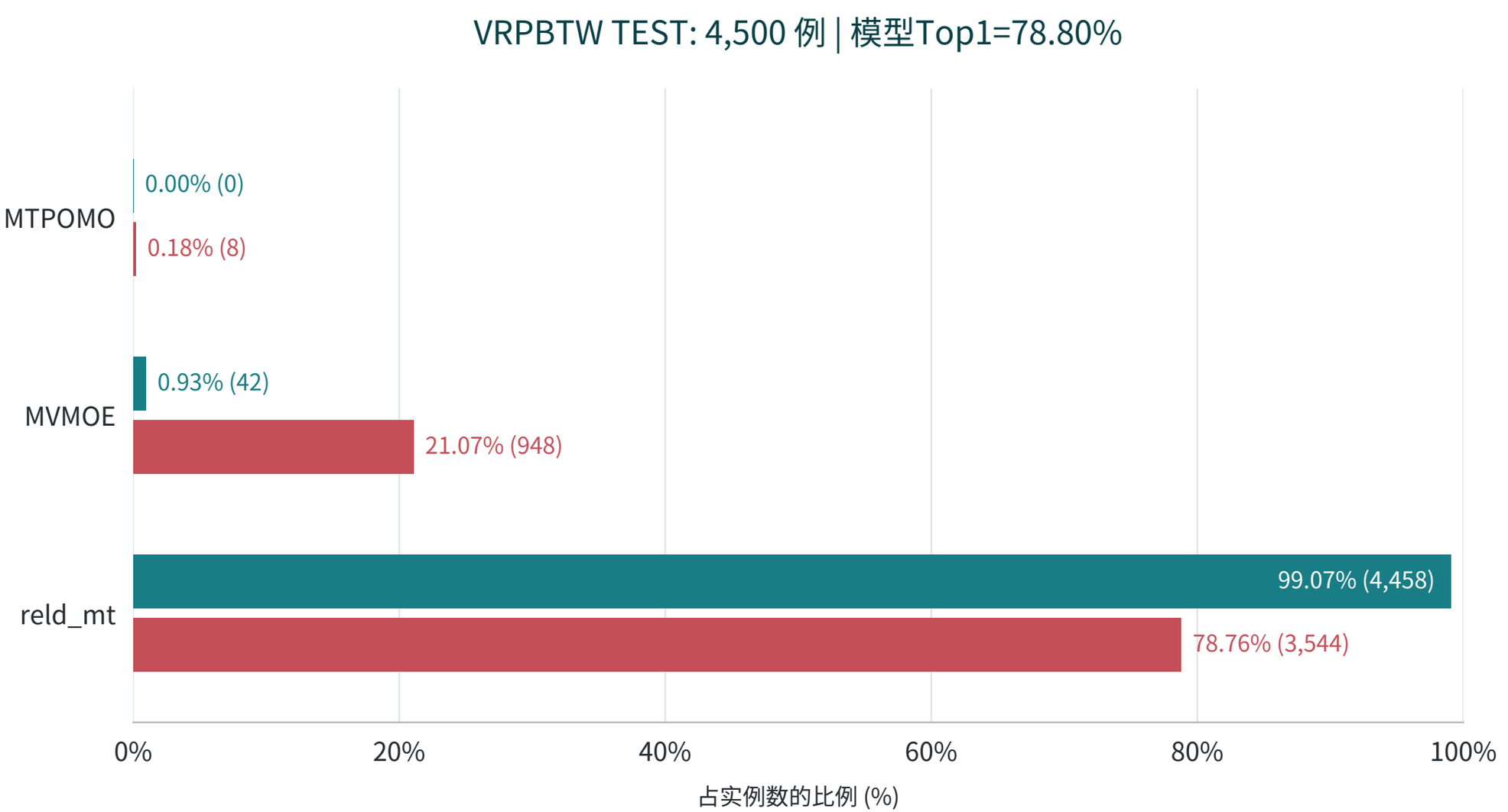
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPBTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

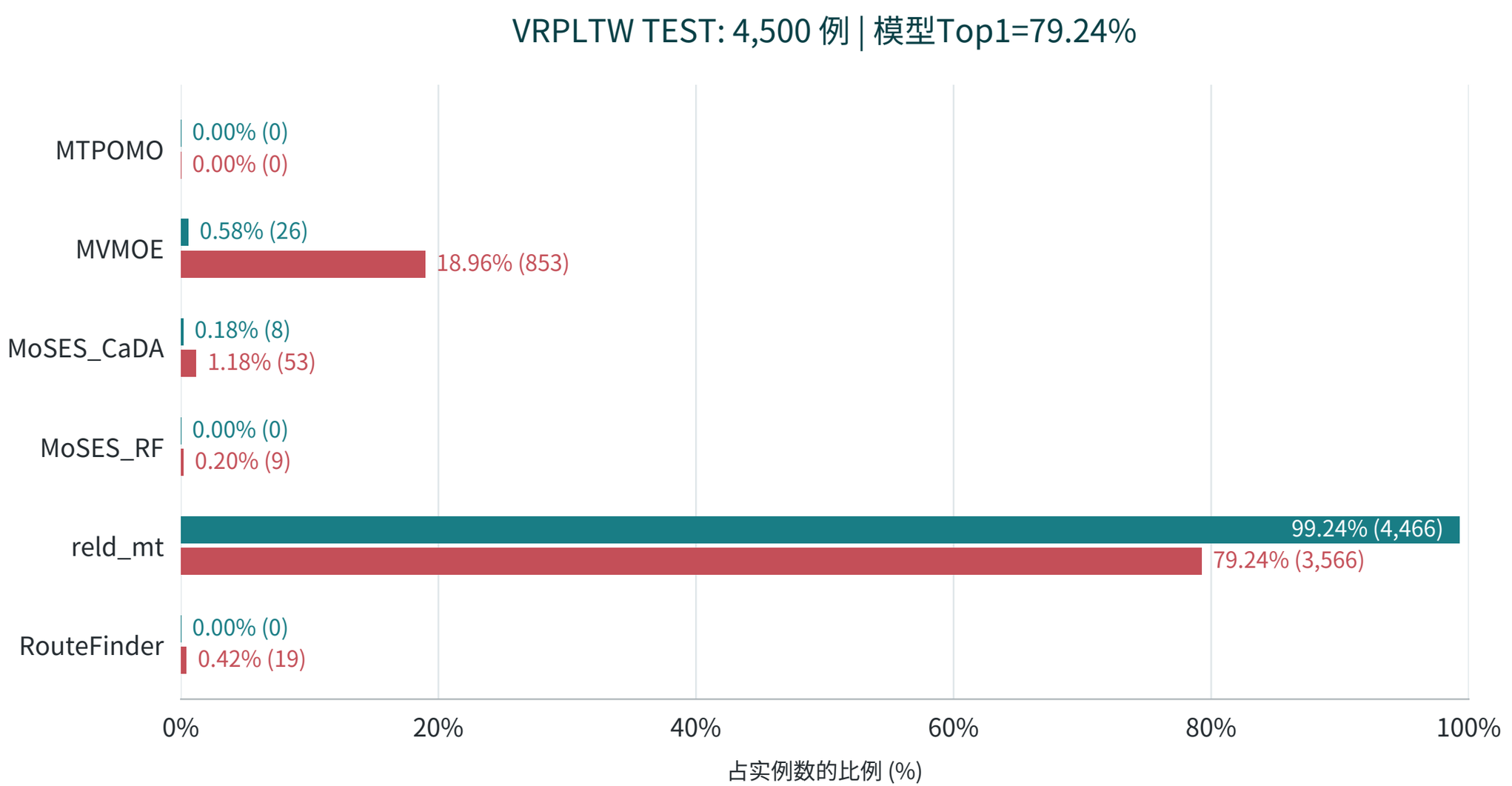
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | VRPLTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

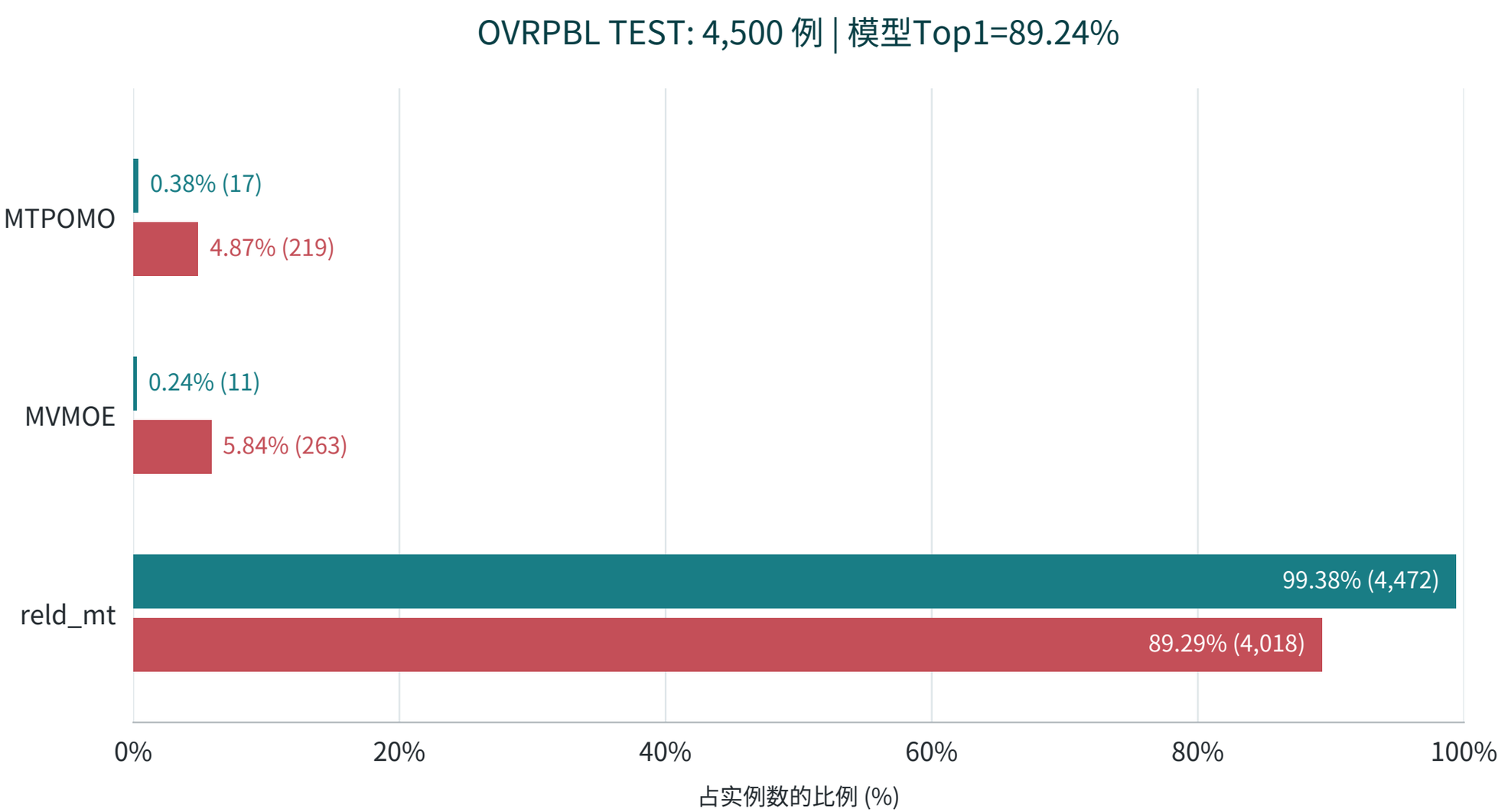
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPBL | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

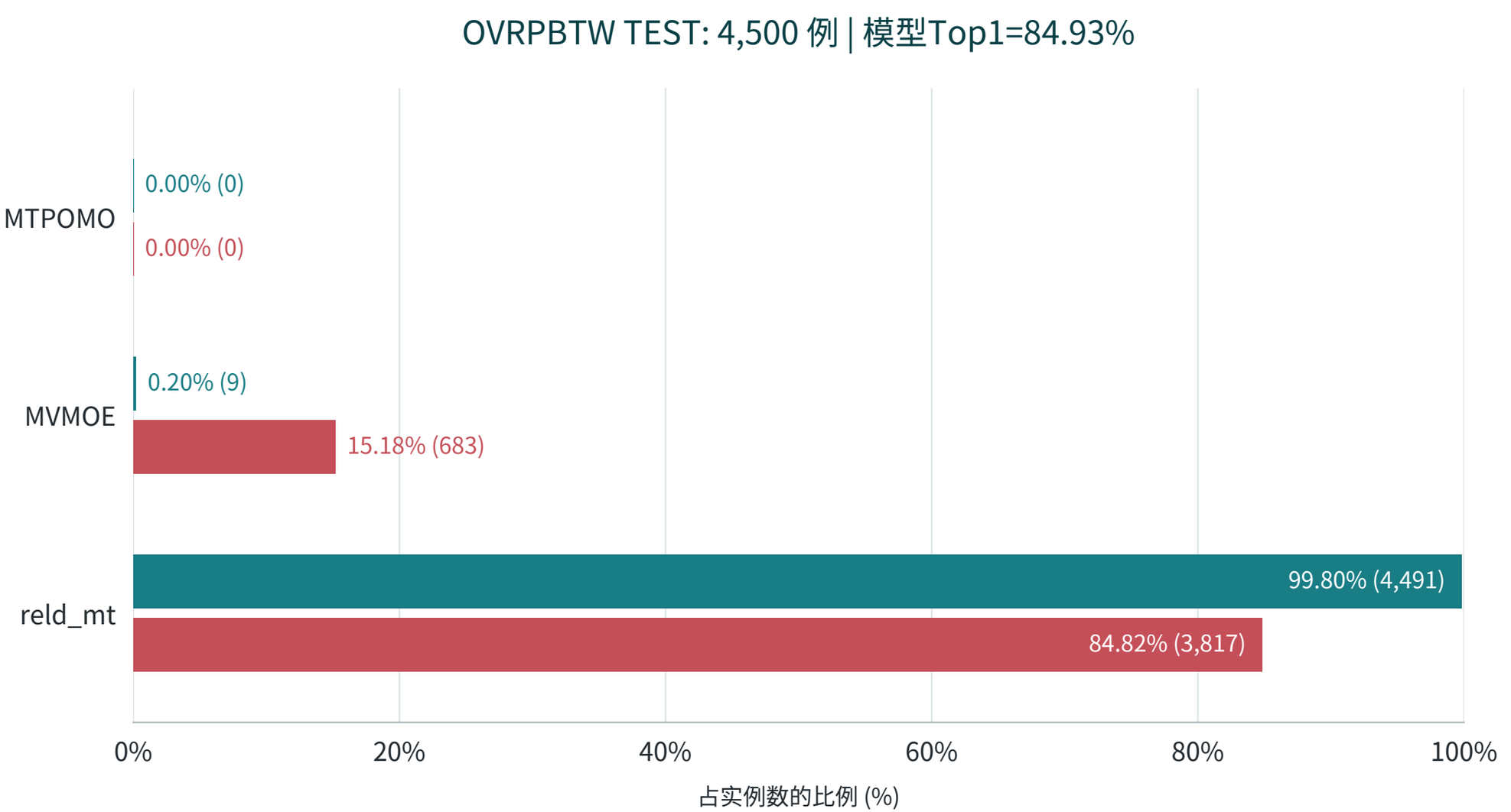
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPBTW | 方法分布

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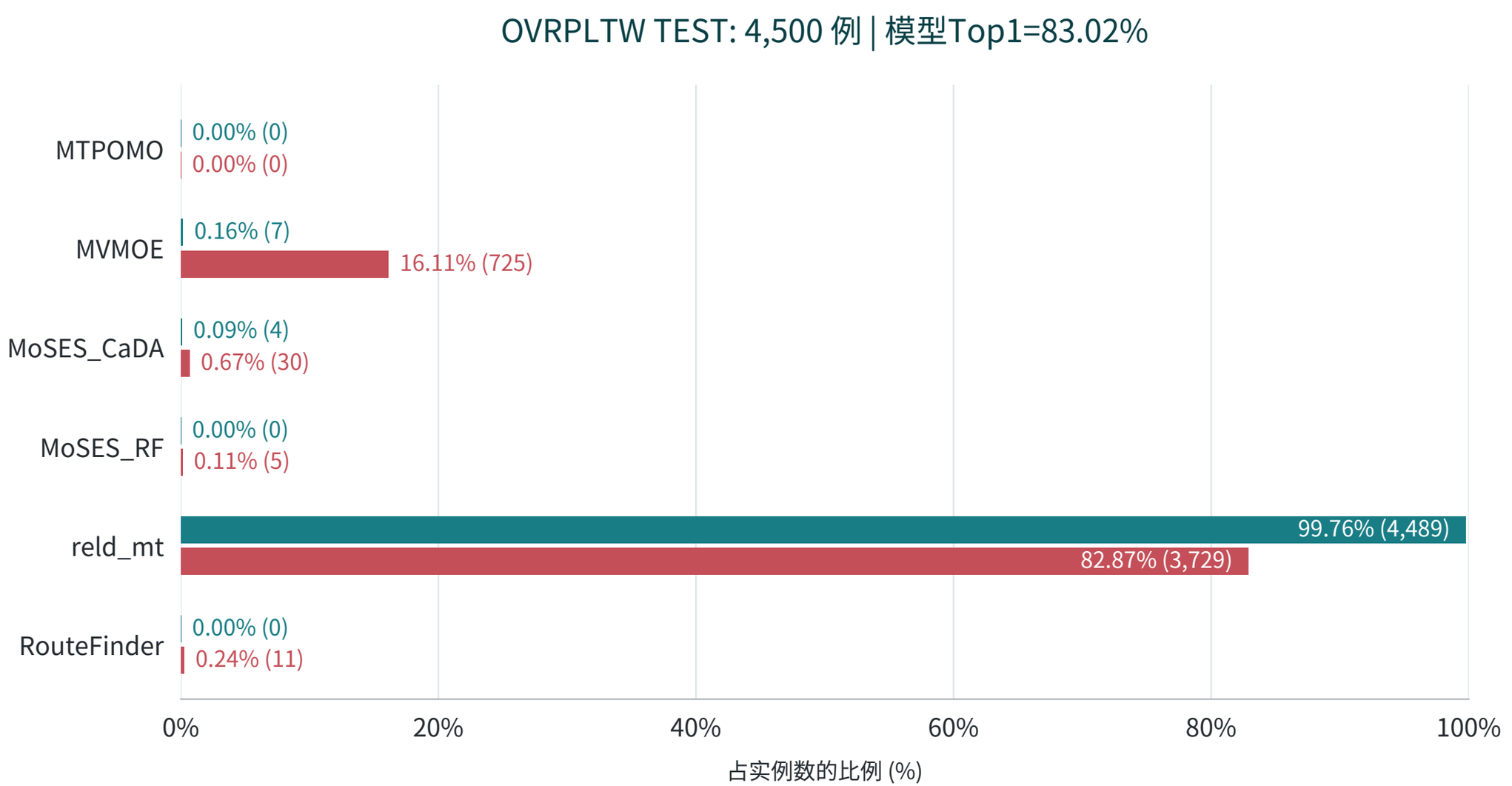
R84/D_attention_focal: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/D_attention_focal | OVRPLTW | 方法分布

同一锁定模型: epoch 30 / update 43200 / SHA256 addbfaff33b5720d

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

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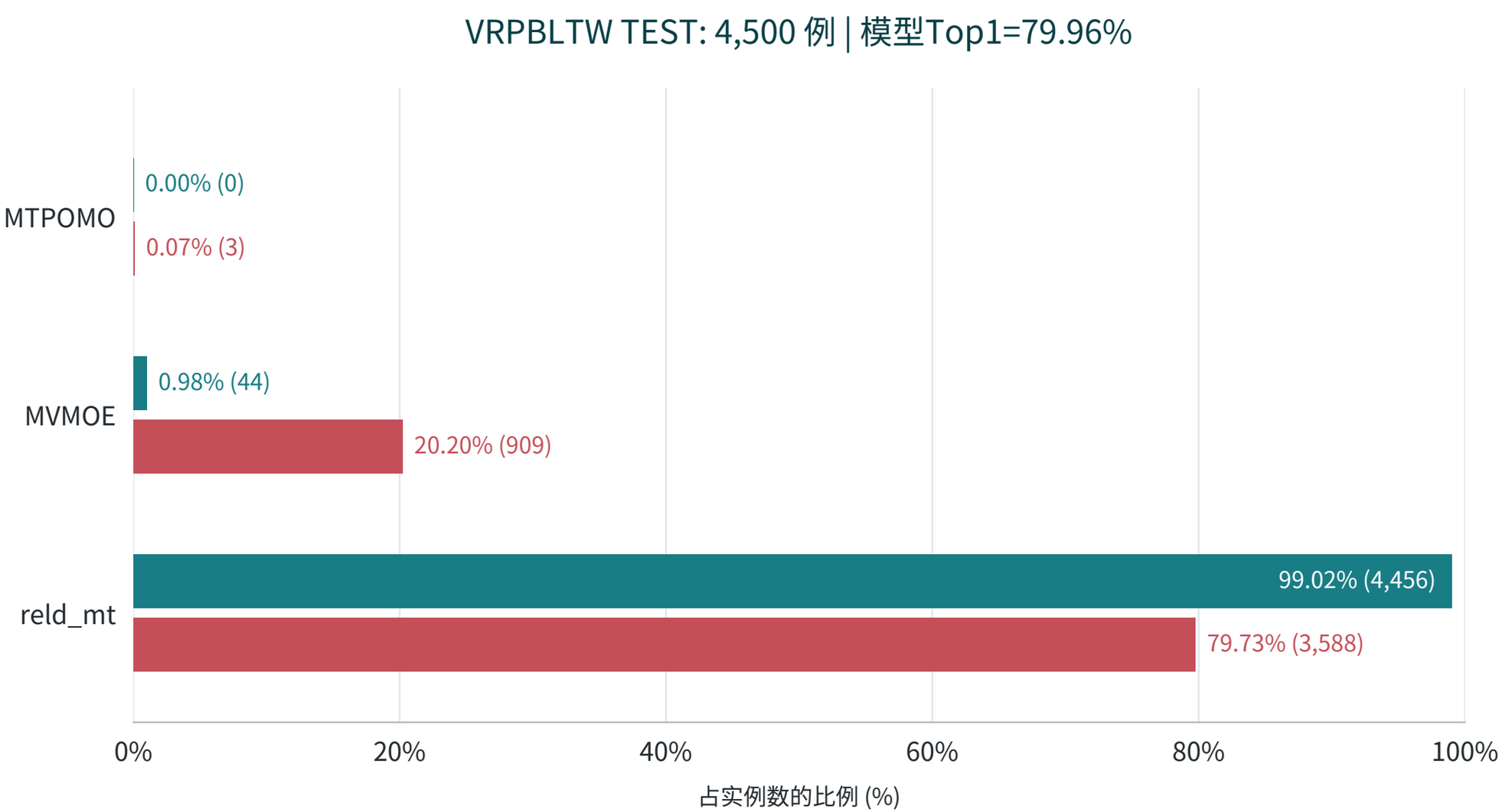
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R84/D_attention_focal | VRPBLTW | 方法分布

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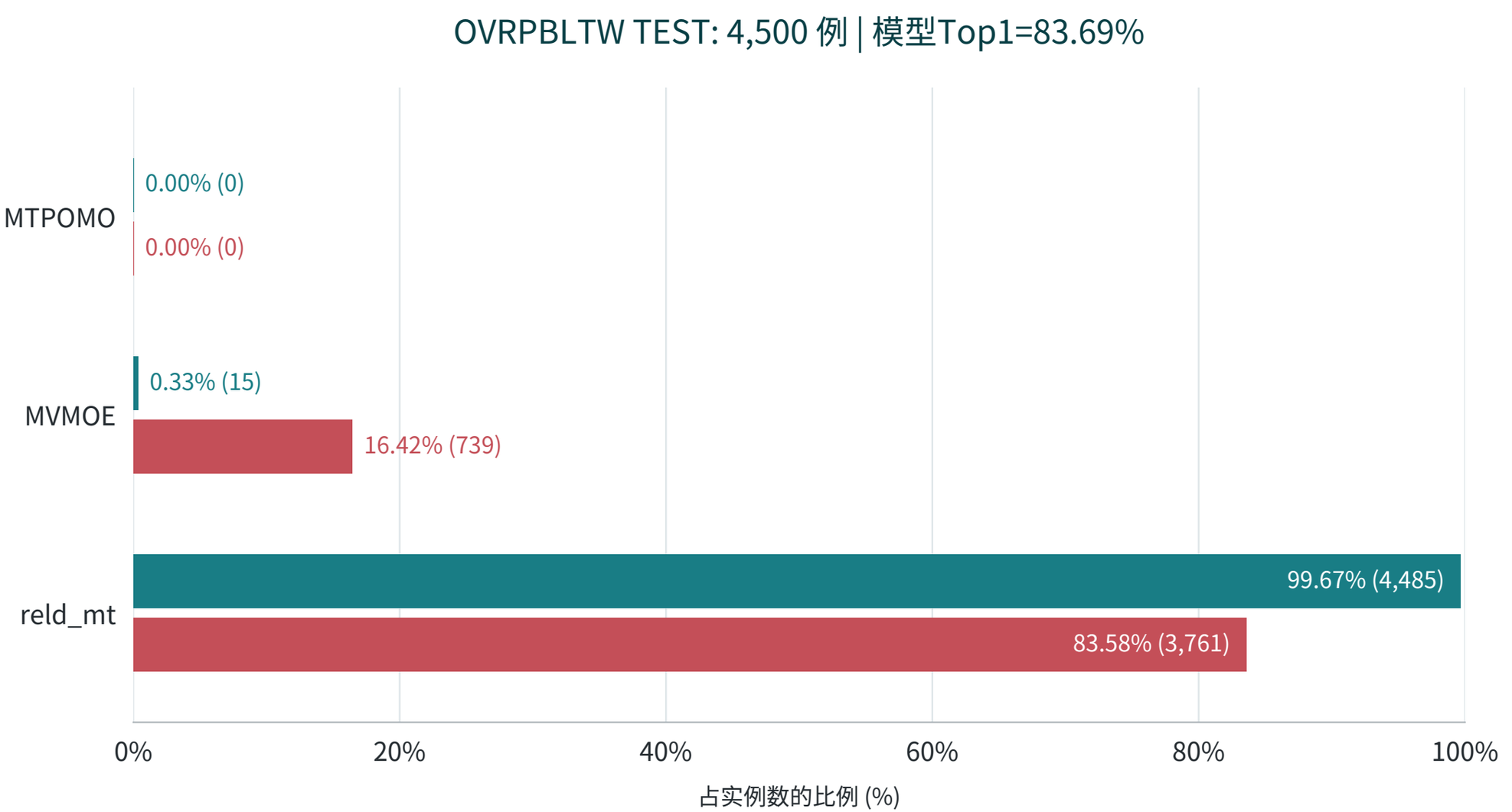
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